LESSON 12.5
FACTORING COMMON MONOMIAL FACTORS

LESSON INTRODUCTION

MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

LEARNING TARGETS

- I can rewrite an algebraic expression with two terms that have a common monomial factor using the distributive property.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

WORKING TOWARD

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.

ESSENTIAL QUESTIONS

- How do I find a common monomial factor?
- How do I factor an algebraic expression that has a common monomial factor?

LESSON NARRATIVE

This lesson builds on student knowledge of rewriting algebraic expressions with a common factor using the distributive property. In this lesson, students extend their understanding of factoring to rewrite algebraic expression with two terms that have a common monomial factor.

COMPONENT SUMMARY

12.5.1 WARM-UP *(~5 MIN.)*

Describe and complete a pattern

12.5.2 EXPLORATION *(10-15 MIN.)*

Explore finding common factors of monomials

12.5.3 BRING IT TOGETHER *(10-15 MIN.)*

Find common factors of monomials

12.5.4 GUIDED PRACTICE *(10-15 MIN.)*

Rewrite expressions using a common monomial factor

12.5.5 COLLABORATION *(10-15 MIN.)*

Analyze student work to determine if errors were made in rewriting sums of algebraic expressions

12.5.6 WRAP-UP *(~5 MIN.)*

Reflect on the lesson

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly apply the laws of exponents. For example, students might write $x \cdot x$ as $2x$ instead of x^2 and $x^2 \cdot x$ as $3x$ instead of x^3.
- Students may believe they have found the Greatest Common Factor by identifying the integer that is a factor. Reinforce for students they should be including variables that the terms also have in common when looking for a monomial factor. For example, students might identify the monomial factor as 8 instead of $8ab^2$. If needed, have students revisit their definition of monomial.

THINGS TO CONSIDER

- Consider strategic pairing of student partners, specifically if students were absent for Lesson 4 and/or if students demonstrated they needed more support by the end of Lesson 4.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

After 12.5.2 Exploration, students can be engaged in the Compare and Connect literacy routine. Ask students to compare and make connections between rewriting algebraic expressions with a common factor using the distributive property, and rewriting algebraic expressions with two terms that have a common monomial factor utilizing the distributive property. Have students share out to the class the comparisons and any helpful connections students mention.

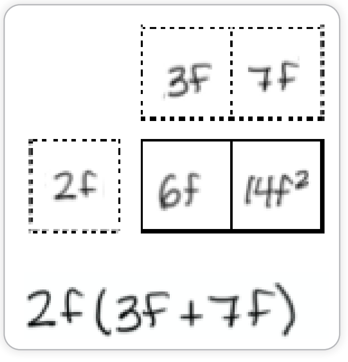
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

12.5.2 Exploration students investigate finding common factors of monomials using the area model. Through repeated practice throughout, the students are able to express the connection between the concept of finding common factors of monomials and the representation.

DIFFERENTIATE INSTRUCTION

Area models provide a visual entry point for students as they consider how to rewrite expressions using monomials. Consider having students explain how using the area model is helpful when rewriting expressions. This can break down a barrier for students who need the representation for support but are afraid to use it when it's not required in front of their peers.



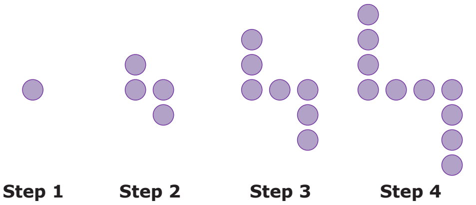
12.5.1 WARM-UP: VISUAL PATTERNS

QUESTIONING

For students who are ready for a challenge, ask them if they can write an algebraic expression that represents the pattern. Ask them to use the rule they create to find the number of circles that will be in step 42.

12.5.1 Warm-Up: Visual Patterns

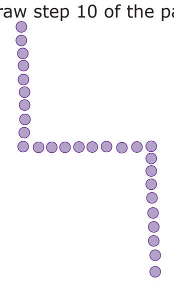
1. The first four steps in a pattern are shown.



   a. Describe how the pattern is changing in each step.
   Answers will vary. Three circles are added each time. One circle is added on the left side, one circle is added on the right side, and one circle is added to the middle.

   b. How many purple circles will be in step 5?
   13

   c. Draw step 10 of the pattern.



12.5.2 Exploration: Finding Common Factors of Monomials

1. Consider the monomial $6z^2$.

   a. The area model below shows two possible factors of $6z^2$.

	z
$6z$	$6z^2$

   b. Complete the equation by filling in the blanks with the factors from the area model.

   $6z^2 = 6z \cdot z$

   c. Work with your partner to identify two other possible pairs of factors for the monomial $6z^2$.
   Answers will vary depending on the common factor chosen.

	z^2
6	$6z^2$

   $6z^2 = 6 \cdot z^2$

	$2z$
$3z$	$6z^2$

   $6z^2 = 3z \cdot 2z$

   d. Explain what strategy you used to identify factor pairs that multiply to get $6z^2$.
   Answers will vary. Find factors of 6 and factors of z^2.

2. Consider the expression $4x^2 + 12x$.

   a. Work with your partner to determine which of the terms shown are common factors of $4x^2$ and $12x$.

   ☐ 4 ☐ x^2 ☐ $4x$ ☐ $2x$

   ☐ $6x$ ☐ 2 ☐ 6 ☐ x

   b. Explain how you made your choices.
   Answers will vary. I chose the common factors of 4 and 12, common factors of x and x^2, and products of those common factors.

   c. Choose one of the common factors from Part A. Work with your partner to complete the area model using the common factor.
   Answers will vary depending on the common factor chosen.

	x	3
$4x$	$4x^2$	$12x$

   Common Factor

   d. Rewrite the expression $4x^2 + 12x$ in factored form using the factors determined in Part C.
   $4x(x + 3)$

   The expression $4x^2 + 12x$ can be re-written using other common factors.

   e. Work with your partner to find three other equivalent expressions using different common factors of $4x^2$ and $12x$.
   $x(4x + 12)$
   $2x(2x + 6)$
   $4(x^2 + 3x)$

12.5.2 EXPLORATION: FINDING COMMON FACTORS OF MONOMIALS

TEACHER MOVES

Assess your student needs to determine if students can be given the option to work in pairs or independently on 12.5.2 Exploration.

TEACHER MOVES

Compare and Connect

Students completed similar problems in 12.4.4 Exploration with rewriting algebraic expressions with a common factor using the distributive property. Have students compare the use of area models and the process to factor these expressions. Make connections between the completion of the area models and the use of the distributive property in the problems.

- What is similar about the problems?
- What is different about the problems?

TEACHER MOVES

Consider calling on students to share their answers to questions 2a and 2b to the class so other students can hear the thinking of their peers.

TEACHER MOVES

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group 12.5.3 Bring It Together. Provide additional scaffolding if needed, but this should be primarily a student-directed exploration.

TEACHER MOVES

Consider having students compare expressions with a different partner group.

Consider asking students to verify their expressions are equivalent by applying distributive property to their rewritten expressions.

12.5.3 BRING IT TOGETHER: FINDING COMMON FACTORS OF MONOMIALS

TEACHER MOVES

Consider having students complete the table independently, then discuss answers with their partner.

12.5.3 Bring It Together: Finding Common Factors of Monomials

Similar to whole numbers, common factors of monomials can be found by considering the factors each term shares.

1. Work with your partner to complete the table. The first row has been completed.

Monomial Terms	Which factor(s) do the coefficients share?	Which variable factor(s) do the terms share?	Common Factors of the Monomials
$15w^2$ and $30w$	3, 5, 15	w	$3w$, $5w$, $15w$
$4c$ and $16c^2$	2, 4	c	$2c, 4c$
$10g^2h^2$ and $20gh^2$	2, 5, 10	g, h, h^2	$2g, 5g, 10g$ $2h, 5h, 10h$ $2h^2, 5h^2, 10h^2$

Common factors of monomial terms can be used to rewrite sums of two algebraic expressions in an equivalent factored form.

TEACHER MOVES

Students may forget to carry the sign with the term and/or incorrectly divide with negative numbers. When giving feedback to students, avoid telling them their errors. Have students verify their models and equivalent expressions are equivalent to -12qr²+20q²r² by using the distributive property.

2. Consider the expression $-12qr^2 + 20q^2r^2$.
   a. Complete the area models in the table to find two equivalent expressions in factored form.

Area Model	Equivalent Expression
Top: $-3r^2$, $5qr^2$ Side: $4q$ Inside: $-12qr^2$, $20q^2r^2$	$4q(-3r^2 + 5qr^2)$
Top: 6, $-10q$ Side: $-2qr^2$ Inside: $-12qr^2$, $20q^2r^2$	$-2qr^2(6 - 10q)$

   b. Explain how the first expression would change if $-4q$ had been used as the common factor instead of $4q$.
   Answers will vary. The signs of the terms inside the parentheses would change. The equivalent expression would be $-4q(3r^2 - 5qr^2)$.

12.5.4 Guided Practice: Rewriting Expressions Using a Common Monomial Factor

1. Use the area model to find an equivalent expression to $32ab^2 + 8a^2b^2$ in factored form.

<table>
<tr><th>Area Model</th><th>Equivalent Expression</th></tr>
<tr><td><table><tr><td></td><td>4</td><td>a</td></tr><tr><td>$8ab^2$</td><td>$32ab^2$</td><td>$8a^2b^2$</td></tr></table></td><td>$8ab^2(4 + a)$</td></tr>
</table>

2. Consider the expression $14jk^2 - 35j^2k^2$.

   a. Complete the area models in the table to find two equivalent expressions in factored form.

<table>
<tr><th>Area Model</th><th>Equivalent Expression</th></tr>
<tr><td><table><tr><td></td><td>2</td><td>$-5j$</td></tr><tr><td>$7jk^2$</td><td>$14jk^2$</td><td>$-35j^2k^2$</td></tr></table></td><td>$7jk^2(2 - 5j)$</td></tr>
<tr><td><table><tr><td></td><td>$2k$</td><td>$-5jk$</td></tr><tr><td>$7jk$</td><td>$14jk^2$</td><td>$-35j^2k^2$</td></tr></table></td><td>$7jk(2k - 5jk)$</td></tr>
</table>

   b. Verify that the expressions are equivalent to $14jk^2 - 35j^2k^2$ using the distributive property.

12.5.4 GUIDED PRACTICE: REWRITING EXPRESSIONS USING A COMMON MONOMIAL FACTOR

ENRICHMENT

Encourage students to synthesize understanding through questioning:

- How did you decide what to use as the common factor?
- Can you think of more ways we could have factored this expression?
- Of the ways you chose to factor $14jk^2 - 35j^2k^2$ which did you find to be a more efficient process? Why?

12.5.5 COLLABORATION: ERROR ANALYSIS

QUESTIONING

Can questions 1-3 be factored differently? Consider having students find another way to factor each expression. Conduct a Gallery Walk for groups to share their new equivalent expressions.

TEACHER MOVES

Students may not recognize the common factor of 1 in question 2 as it has not appeared previously in the lesson. If needed, partner students or bring the class together based on the number of students struggling with understanding where the 1 came from.

12.5.5 Collaboration: Error Analysis

One student's work rewriting sums of algebraic expressions using common factors is shown in the table. Work with your partner to analyze the student's work and complete the table.

Original Problem and Student Work	Are there any errors?	If there was an error, circle the error in the original work and write a correction.
1. Write an equivalent expression to $6f + 14f^2$. $3f$ $7f$ $2f$ $6f$ $14f^2$ $2f(3f+7f)$	○ **Yes** ○ No	$2f \cdot 3 = 6f$ $6f + 14f^2 = 2f(3 + 7f)$
2. Write an equivalent expression to $3m^2n + 15m^2n^2$. 1 $5n$ $3m^2n$ $3m^2n$ $15m^2n^2$ $3m^2n(1+5n)$	○ Yes ○ **No**	
3. Write an equivalent expression to $-25jk^2 + 45jk$. $-5k$ $9jk$ $5jk$ $-25jk^2$ $45jk$ $5jk(-5k+9jk)$	○ **Yes** ○ No	$5jk \cdot 9 = 45jk$ $-25jk^2 + 45jk = 5jk(-5k + 9)$

12.5.6 Wrap-Up: Growth Mindset

Complete each statement based on today's lesson.

1. Today I am still unsure about …
   Answers will vary.

2. I will try to build my confidence by …
   Answers will vary.

12.5.6 WRAP-UP: GROWTH MINDSET

TEACHER MOVES

Consider pairing students to discuss their reflections to see if peers can answer any of their questions.

DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.